



Sea ice

In the polar regions, vast areas of ocean are covered with floating ice, especially in winter. The ice forms as freezing winds chill the ocean surface, creating ice crystals that fuse together into solid ice sheets that can be several feet thick. The sea ice may be frozen to the shore and covered with snow, so it looks like an extension of the land. But most of the floating ice is not attached to the shore, and drifts with the currents as pack ice.



WHY ICE FLOATS

When water freezes, its triangular molecules lock together in a three-dimensional structure with open space between them. In the process, the molecules move apart, so a liter of ice contains fewer molecules than a liter of water, and therefore weighs less. This is why ice floats. No other substance behaves like this; it is a unique property of water.

◀ PENGUIN PERCH

If ice behaved like the solid forms of all other liquids, it would sink to the seabed—and these penguins would have nowhere to rest.

FREEZING SEAS

Sea ice does not form in a solid sheet. If the air temperature keeps falling, the water freezes in stages starting with a mass of small ice crystals called frazil or grease ice. These crystals freeze together to form roughly circular plates of pancake ice. They then fuse into thick pack ice that breaks up and freezes together again many times, but eventually becomes welded together into a solid sheet.



▲ GREASE ICE

Ice crystals freezing at the surface form a layer of slushy ice, like liquid mud; seals can surface through it.



▲ PANCAKE ICE

The grease ice forms plates, and as these bump together their edges turn up, like pancakes.



▲ MULTI-YEAR ICE

The ice forms a rough, tumbled sheet made up of thick ice floes pushed together by the wind.